HEIQ XMEMBRANE PROJECT TEAM

Model closure and material handoff

What the integrated ERV+EO model has established

31 August 2026 · Revision 2

Controlled correction to the earlier closure note

This revision uses the selected architecture only: EO is integrated in the ERV membrane/core. It supersedes the earlier closure report wherever that report described a downstream or post-ERV EO module, used the historical 180-Wh/kg shorthand as break-even, or attributed an AC-only difference to EO.

Closure in one sentence. The equations, signed transport audit and matched architecture comparison are internally consistent and identify an O30 opportunity below 9,000 m^2; the remaining uncertainty is not another algebraic degree of freedom, but measured material coupling and the omitted receiver, heat-rejection, conversion and hardware power.

Fixed service	O80 at 32 °C/80% RH to 20 °C/50% RH supply
Selected point	Integrated ERV+EO to O30; 5,899 m^2; 418.400 kW known subtotal
Passive control	Same ERV with EO off plus AC: 543.741 kW
Incremental window	125.340 kW or 23.05% versus the passive control
Calculated break-even	51.955 Wh/kg all-in active-core energy
Claim boundary	Conditional model window; no validated membrane, module or net-system saving

1 What “model closure” means

THE MODEL QUESTION IS CLOSED; THE MATERIAL QUESTION IS DELIBERATELY OPEN

Model closure does not mean that a product has been validated. It means that the previously ambiguous definitions, signs, comparators and decision coordinates have been made explicit enough that the model gives reproducible answers and exposes the measurements that now control the decision.

Question	Closed result	What remains open
Where EO acts	Inside the ERV membrane/core only	Core construction and current collection
What O50/O30 mean	Intermediate humidity-ratio targets, not supply-air states	Measured achievable outlet under real films
What EO is compared with	Same passive ERV+AC system with EO off	Full-system parasitic ledger
Whether O50 needs EO	No; passive ERV already reaches 13.215 g/kg, below O50 at 14.952 g/kg	Sensitivity to measured exchanger performance
Whether O30 fits the area gate	Yes in the screened model: 5,899 m^2 at 1.00 V	Manufacturable packing and pressure drop
Whether full EO drying fits	No; the minimum valid screened full-duty area is about 17,253 m^2	No further claim below 9,000 m^2

The thermodynamic audit also closes the sign convention. A locally positive passive surface direction is not a rejection criterion for electrically driven transport against that passive tendency. Feasibility is tested with the total electrochemical driving force and entropy-production sign, not by requiring the passive water-activity gradient to point in the desired pumping direction.

> The phrase “no primary row resolves the local surface direction” meant that the screened literature or property rows did not independently determine the sign and magnitude of the membrane-surface water-activity gradient under the actual ERV film conditions. That absence prevents material ranking from that single gradient; it does not invalidate active uphill EO transport.

2 Energy closure for the integrated architecture

THE BREAK-EVEN IS CALCULATED FROM THE MATCHED PASSIVE CONTROL

For the selected O30 point, the integrated core removes 4,365.918 kg/h of water. The remaining AC load is 316.910 kW. The maximum all-in active-core specific energy that preserves parity with the same passive ERV+AC control is therefore

$$e_{\mathrm{BE}} = \frac{543.741 - 316.910}{4{,}365.918/3{,}600} = 51.955\ \mathrm{Wh/kg}.$$

The known active-core term is 101.490 kW, or 23.246 Wh/kg. The difference is 28.709 Wh/kg, equivalent to 125.340 kW. This is the complete remaining budget for the loads omitted from the present subtotal.

> **No net-saving claim yet.** Receiver work, heat rejection, conversion losses, electrodes, current collection, manifold losses, incremental fan power, leakage and durability are not all closed. Their combined O30 load must remain below 125.340 kW for the conditional incremental advantage to survive.

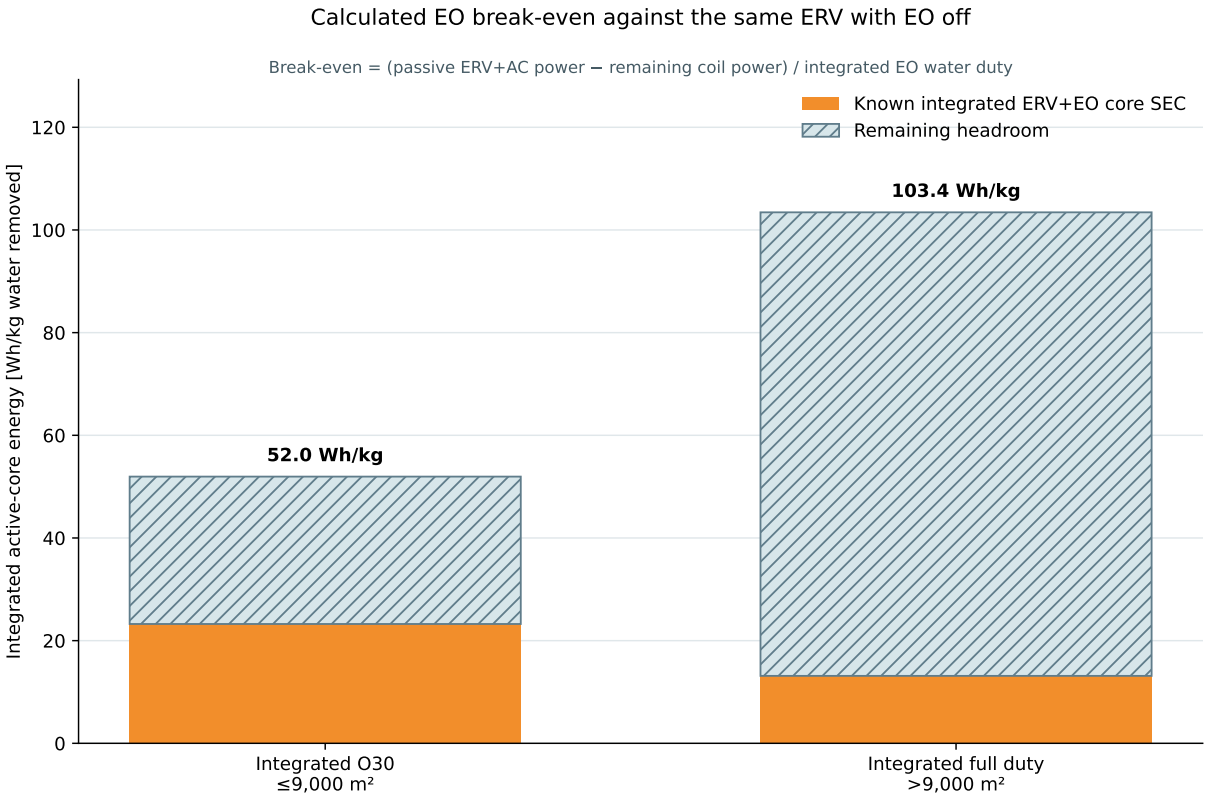


Figure 1: Architecture-specific break-even. The historical 180-Wh/kg value is not used as the selected break-even.

3 What was learned about membrane selection

THERE IS NO VALID SINGLE-PROPERTY WINNER

The model rejects selection by water diffusivity alone. The EO response and power depend on a coupled property bundle evaluated at the same temperature, water activity, mechanical compression and electrode state:

$$\left\{ \frac{D_w}{L}\frac{\partial C_w}{\partial a_w},\ n_{\mathrm{drag}},\ \sigma_{\mathrm{wet}},\ ASR_{\mathrm{compressed}},\ L,\ C_w(a_w),\ \text{mechanical stability} \right\}.$$

High sorption or diffusivity may improve passive permeation while low wet conductivity or high compressed resistance raises electrical loss. Likewise, large electro-osmotic drag is not useful if current must pass through an unacceptably resistive wet membrane. The properties must therefore be measured as a joint vector on the same sample, not assembled from unrelated best values.

3.1 Symmetry and Yaroshchuk-type asymmetry

The current model includes a symmetric quasi-static $+J/0/-J$ replay. Under the present constant-drag constitutive law, the positive- and negative-current contributions cancel around the passive state, producing no useful mean EO pumping. The study has **not** yet implemented a validated asymmetric Yaroshchuk-style membrane, rectifying interfaces, direction-dependent drag or memory. Such an effect could change the cycle mean, but only after its constitutive relation and parameters are supplied by measurement or a traceable source.

Material handoff requirement. Measure on the same conditioned coupon: sorption isotherm and derivative, water mobility/permeance, EO drag versus current and polarity, wet conductivity, compressed ASR, thickness, swelling, gas permeability and mechanical response. Preserve raw polarity-resolved transients so asymmetry can be tested rather than assumed.

4 Experimental closure route

A CONTROLLED-FLOW COUPON CELL CAN FALSIFY THE MODEL ASSUMPTIONS

The recommended first fixture is a two-sided controlled-flow vapour cell with a 50×50-mm active aperture (25 cm^2), approximately 80×80-mm coupon and 110×110-mm outer plates. Use interchangeable symmetric 1/2/4-mm flow gaps, with 2 mm nominal. At 1 m/s the nominal actual flow is 6.0 L/min per side. Start with 8–10-mm insulating flow frames and 12-mm metal backing/load plates; a load-bearing PEEK plate should start near 16 mm and be accepted only after loaded flatness and gap mapping.

The required controls are a split common inlet, EO-off/EO-on ABBA sequencing, two-sided water balance, foil blank, polarity reversal, pressure control, gap/flow repeats, reference cross-switching and multiple perimeter inserts. Closed chambers are acceptable only as transient inventory tests; they must not be interpreted as fixed-boundary steady-state measurements.

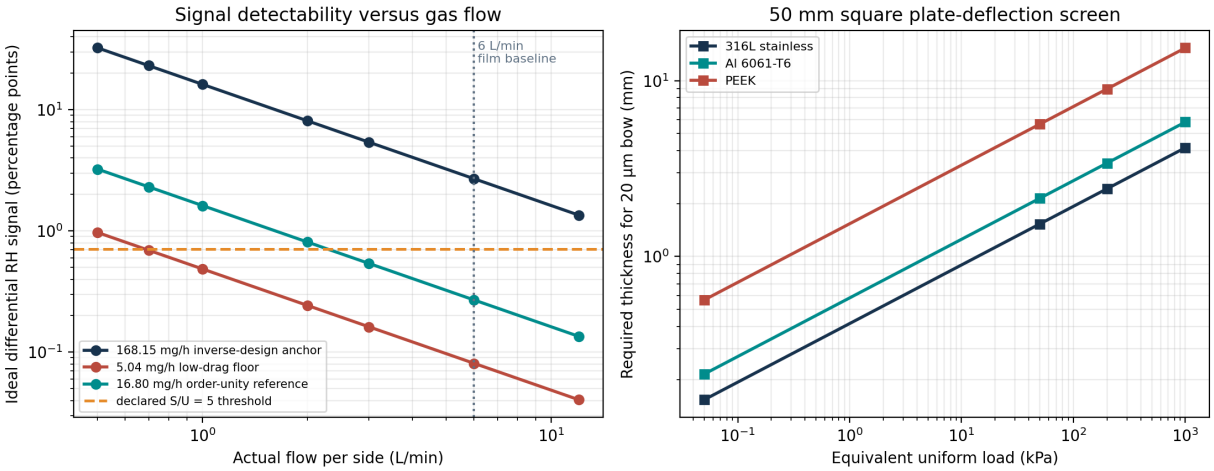


Figure 2: Coupon-cell sizing and detectability trade-offs for the recommended starting fixture.

5 Decision and completion criteria

THE CORRECT NEXT STEP IS EVIDENCE GENERATION, NOT ANOTHER UNCONSTRAINED SWEEP

Decision	Status	Reason
Use passive ERV as the primary control	GO	Established matched reduction to 543.741 kW
Test EO integrated in ERV to O30	GO	5,899 m^2; 125.340-kW conditional window
Assign EO duty to O50	NO DUTY	Passive ERV already passes O50
Claim full EO drying below 9,000 m^2	NO-GO	Minimum valid screened area is about 17,253 m^2
Claim measured net energy saving	NOT YET	Omitted subsystem ledger remains open
Select a commercial membrane	NOT YET	Same-sample coupled properties are unmeasured
Build the controlled-flow coupon fixture	GO	Dimensions, controls and error mechanisms are defined

Model-phase conclusion. The model is sufficiently closed to stop unfocused parameter exploration and proceed to a matched EO-off/EO-on coupon campaign. The O30 opportunity survives only if measured active-core behaviour and the complete omitted-load ledger remain below the calculated 51.955-Wh/kg break-even. The next decision is empirical, not cosmetic.

Controlled evidence basis

This report is subordinate to the Version 9 final result and decision report. Its numerical evidence is reproduced from the integrated active-core case register, the under-9,000-m^2 selection register, the matched opportunity plots and the vapour-cell sizing and boundary-condition analyses included in the controlled Version 9 release package.

End of Model Closure and Material Handoff, Revision 2. This document contains conditional modelling and experiment-planning results; it is not a product-performance certificate.